Post Intubation

Use this guideline if the patient has been intubated and ETT placement has been confirmed by capnography. If the patient has been intubated during cardiac arrest, this guideline should only be used if sustained ROSC occurs.

- Ventilate using the **'mechanical ventilation'** guideline.
- Administer fentanyl and ketamine:
  - Draw up 200 mcg of fentanyl in combination with 200 mg of ketamine. Dilute with 0.9% sodium chloride to a total of 20 ml. This solution contains 10 mcg/ml of fentanyl and 10 mg/ml of ketamine.
  - Commence an infusion at 0.2 ml/kg/hr using an infusion device, or
  - For an adult administer 5 ml of this solution IV (50 mcg of fentanyl and 50 mg of ketamine) every 15-30 minutes, titrating the frequency to the clinical signs of the patient's level of sedation.
  - For a child administer the volume described in the paediatric drug dose tables IV every 15-30 minutes, titrating the frequency to the clinical signs of the patient's level of sedation.
- Administer rocuronium as required:
  - 100 mg IV for an adult weighing greater than 80 kg.
  - 50 mg IV for an adult weighing 50-80 kg.
  - See the paediatric drug dose tables for a child.
- Administer 0.9% sodium chloride IV if there are signs of hypovolaemia or the patient is hypotensive:
  - 500 ml for an adult.
  - 10 ml/kg for a child.
  - Repeat as required.
- Administer metaraminol IV in addition to 0.9% sodium chloride, if the patient remains hypotensive.
  Metaraminol for an adult:
  - Administer via an infusion device:
    - Administer an initial bolus of 0.5-1 mg metaraminol IV.
    - Start the infusion at 2 mg/hour and adjust the rate as required.
  - Alternatively, administer a bolus of 0.5-1 mg metaraminol IV every 5-10 minutes as required.

  Metaraminol for a child:
  - Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.
- Consider administering adrenaline IV if the blood pressure is unresponsive to metaraminol or the patient is hypotensive and bradycardic.
  Adrenaline for an adult:

Post Intubation

- Administer an adrenaline infusion via an infusion device. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
- Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
  - Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

Adrenaline for a child aged 5-14 years:

- Administer an adrenaline infusion via an infusion device. Start at a rate of 0.25 mg/hour and adjust the rate as required, or
- Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - Administer as an infusion. Start at 1 drop/second and adjust the rate as required, or
  - Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

Adrenaline for a child aged less than five years:

- Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.
- Do not administer adrenaline as an IV infusion.

Additional information

Sedation and neuromuscular blockade

- Provided the patient is not in cardiac arrest, all patients that have been intubated should receive sedation (preferably via an infusion device) and neuromuscular blockade.
- Consider administering sedation without neuromuscular blockade, provided this is safe, if it is considered likely that the patient will receive end of life care within the emergency department.
- When administering sedation:
  - Use doses at the lower end of the described ranges if the patient is small, frail, or cardiovascularly unstable.
  - Titrate the dose and frequency of administration to the clinical signs of the patient's level of sedation. Increase the dose and frequency of administration if the patient has hypertension and tachycardia or shows signs of tears. Decrease the dose and frequency of administration if the patient has hypotension.

Ensuring a safe ETT cuff seal

- It is very easy to generate a high pressure within the ETT cuff and this may cause tracheal injury by impairing the blood supply to the tracheal mucosa. Tracheal injury may lead to tracheal stenosis and this may cause significant morbidity.

Post Intubation

- As soon as feasible, the cuff should be checked to ensure that the cuff pressure is between 20-30 cmH_2O.

More Information

Version history

28 October 2024 1.0.5

Update: Multiple changes.